```
Problem 1
Number of Advertisements: [1 2 3 4 5 6]
Monthly Sales: [ 3  5  7 10 12 14]

Part a
Custom Linear Regression Results
Slope: 2.2571
Y-Intercept: 0.6000
Scikit Learn Linear Regression Results
Slope: 2.2571
Y-Intercept: 0.6000

Part b
Sales Predicitons
4 Months: 9.63
8 Months: 18.66

Part c
SST: 89.5000
SSR: 89.1571
SSE: 0.3429
R^2: 0.9962
```

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Problem 2
   Recovery_Time Treatment  Gender
0             10   Placebo    Male
1              8      Drug  Female
2             15   Placebo  Female
3             12      Drug    Male
4              7      Drug    Male
5             14   Placebo  Female
6              9      Drug  Female
7             11   Placebo    Male
8              6      Drug  Female
9             13   Placebo    Male

Part a
   Recovery_Time  Treatment_Placebo  Gender_Male
0             10                  1            1
1              8                  0            0
2             15                  1            0
3             12                  0            1
4              7                  0            1
5             14                  1            0
6              9                  0            0
7             11                  1            1
8              6                  0            0
9             13                  1            1
Scikit Learn Multi Linear Regression Results
Equation: 4.3333x1 + -0.6667x2 + 8.6667

Part b
The coefficients represent the magnitude of the slope and the amount of influence the Treatment or Gender has on the recovery time.
More specifically, the bigger the magnitude, the more influence it has on the recovery time of the patient, in days.

Part c
Recovery Time (Days) Predicitons
Female Patient who Recieved the Drug: 8.6667
Male Patient who Recieved the Drug: 8.0000
```

Problem 3

```
    Price     Size  Bedrooms  Age  Distance_to_City_Center
0  245.72  1896.84         2   10                    25.92
1  349.87  1773.19         4    3                    24.82
2  314.15  2134.47         6   38                    27.39
3  224.69  2980.16         6   34                     4.73
4  271.07  3601.46         3    6                     3.37
```

Part a

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  Price   R-squared:                       0.151
Model:                            OLS   Adj. R-squared:                  0.075
Method:                 Least Squares   F-statistic:                     2.000
Date:                Sat, 14 Feb 2026   Prob (F-statistic):              0.111
Time:                        09:36:17   Log-Likelihood:                -271.14
No. Observations:                  50   AIC:                             552.3
Df Residuals:                      45   BIC:                             561.8
Df Model:                           4
Covariance Type:            nonrobust
===========================================================================================
                              coef    std err          t      P>|t|      [0.025      0.975]
-------------------------------------------------------------------------------------------
Size                       -0.0113      0.010     -1.131      0.264      -0.031       0.009
Bedrooms                   -6.4154      7.170     -0.895      0.376     -20.857       8.026
Age                         1.2287      0.575      2.136      0.038       0.070       2.387
Distance_to_City_Center     0.7193      1.084      0.663      0.510      -1.465       2.903
const                     313.8006     50.456      6.219      0.000     212.178     415.423
==============================================================================
Omnibus:                        1.423   Durbin-Watson:                   2.077
Prob(Omnibus):                  0.491   Jarque-Bera (JB):                1.057
Skew:                          -0.015   Prob(JB):                        0.590
Kurtosis:                       2.288   Cond. No.                     1.55e+04
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.55e+04. This might indicate that there are
strong multicollinearity or other numerical problems.
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Part b

Based on the p-values in the summary table, the only variable that significantly impacts the house price is Age. This is because Age has a p-value of 0.038, which is less than the significance level of 0.05. All other variables (Size, Bedrooms, Distance) have p-values greater than 0.05, meaning they are not statistically significant.

Part c

House Sales Prediction

Parameters: Size; 2500ft, Bedrooms; 4, Age; 20 years, Distance to City Center; 10 miles

Cost: $291.7429